

Linear Algebra

CUSAT M.Tech. Computer Science & Engineering
Artificial Intelligence & Software Engineering

Sayona C. Sajan

Cochin University of Science and Technology

September 2026

Vectors $\rightarrow$ Transformations $\rightarrow$ Subspaces $\rightarrow$ Eigenstructure

Why Linear Algebra matters in AI & Software Engineering

Core idea

Linear algebra gives a language for representing data, transformations and constraints.

- ▶ Data points and features $\rightarrow$ vectors
- ▶ Neural-network layers $\rightarrow$ matrix transformations
- ▶ Similarity $\rightarrow$ dot products
- ▶ Geometry $\rightarrow$ transformations and cross products
- ▶ Stable directions $\rightarrow$ eigenvectors

1. Vectors and coordinate systems

A vector has both **magnitude** and **direction**.

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + v_3 \mathbf{e}_3$$

where

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

- ▶ $\mathbb{R}^2$: ordered pairs (x, y)
- ▶ $\mathbb{R}^3$: ordered triples (x, y, z)
- ▶ The origin is the zero vector.
- ▶ Coordinates depend on the chosen basis.

Vector addition and scalar multiplication

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix},$$

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{bmatrix}, \quad c\mathbf{u} = \begin{bmatrix} cu_1 \\ cu_2 \\ cu_3 \end{bmatrix}.$$

Example

$$\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \\ 7 \end{bmatrix}.$$

Think geometrically: addition translates one arrow; scalar multiplication stretches, shrinks, or reverses it.

Linear combinations

A **linear combination** of $v_1, \dots, v_k$ is

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k.$$

Key question

Can a target vector b be written as a linear combination of the given vectors?

$$Ax = b$$

is exactly the coefficient-finding problem.

Example

$$2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}.$$

Thus every vector in $\mathbb{R}^2$ is a linear combination of the standard basis vectors.

2. Span

The **span** of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is the set

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \left\{ \sum_{i=1}^k c_i \mathbf{v}_i : c_i \in \mathbb{R} \right\}.$$

- ▶ One nonzero vector in $\mathbb{R}^2$ spans a line through the origin.
- ▶ Two independent vectors in $\mathbb{R}^2$ span the whole plane.
- ▶ Three independent vectors in $\mathbb{R}^3$ span all of $\mathbb{R}^3$.

Basis and dimension

A set $B = \{v_1, \dots, v_n\}$ is a **basis** if it is:

- 1 spanning, and
- 2 linearly independent.

The number of basis vectors is the **dimension**.

$$\dim(\mathbb{R}^n) = n.$$

Standard basis of $\mathbb{R}^3$

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Important equivalence in $\mathbb{R}^n$

For n vectors: independent $\iff$ spanning $\iff$ basis.

3. Matrix multiplication as composition

If

$$T(\mathbf{x}) = A\mathbf{x}, \quad S(\mathbf{x}) = B\mathbf{x},$$

then applying T first and S second gives

$$S(T(\mathbf{x})) = B(A\mathbf{x}) = (BA)\mathbf{x}.$$

Order matters

$$BA \neq AB \quad \text{in general.}$$

Matrix multiplication represents composition of transformations, so the rightmost matrix acts first.

Mental model

$$\mathbf{x} \xrightarrow{A} A\mathbf{x} \xrightarrow{B} BA\mathbf{x}.$$

Matrix multiplication: row $\times$ column

For compatible matrices A and B ,

$$(AB)_{ij} = \sum_k a_{ik} b_{kj}.$$

Example

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1(5) + 2(7) & 1(6) + 2(8) \\ 3(5) + 4(7) & 3(6) + 4(8) \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

Dimension rule

$$(m \times n)(n \times p) = (m \times p).$$

Three-dimensional linear transformations

A linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ satisfies

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}), \quad T(c\mathbf{u}) = cT(\mathbf{u}).$$

A 3×3 matrix acts on $\mathbb{R}^3$.

- ▶ Scaling
- ▶ Reflection
- ▶ Rotation
- ▶ Shearing
- ▶ Projection

Columns reveal the map

If $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3]$, then

$$A\mathbf{e}_1 = \mathbf{a}_1, \quad A\mathbf{e}_2 = \mathbf{a}_2, \quad A\mathbf{e}_3 = \mathbf{a}_3.$$

Common 3D transformation matrices

Scaling

$$S = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix}$$

Rotation about z

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection in xy -plane

$$F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Projection onto xy -plane

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

4. Determinant

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \det(A) = ad - bc.$$

Geometrically, $|\det A|$ is the **area/volume scaling factor**.

- ▶ $\det(A) > 0$: orientation preserved.
- ▶ $\det(A) < 0$: orientation reversed.
- ▶ $\det(A) = 0$: space is collapsed into a lower dimension.

Important

$$\det(A) = 0 \iff A \text{ is singular.}$$

Determinant of a 3×3 matrix

For

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg).$$

Fast method

Expand along a row/column with zeros whenever possible. For hand calculations, keep the alternating signs:

$$+ \quad - \quad +.$$

Transformation viewpoint

If $|\det A| = 3$, every volume is multiplied by 3 under A .

Inverse matrices

An inverse A^{-1} satisfies

$$AA^{-1} = A^{-1}A = I.$$

For a 2×2 matrix,

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \quad \det(A) \neq 0.$$

- ▶ A^{-1} exists iff $\det(A) \neq 0$ for a square matrix.
- ▶ Solving $Ax = b$:

$$x = A^{-1}b.$$

- ▶ Numerically, Gaussian elimination is usually preferable to explicitly computing A^{-1} .

5. Column space

For $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$,

$$\text{Col}(A) = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}.$$

It is the set of all possible outputs:

$$\text{Col}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}.$$

Connection to systems

$A\mathbf{x} = \mathbf{b}$ has a solution exactly when

$$\mathbf{b} \in \text{Col}(A).$$

Null space

The null space is

$$\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}.$$

It contains every input that gets mapped to zero.

Example

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

Then

$$x + 2y = 0 \quad \Rightarrow \quad \mathbf{x} = t \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Hence

$$\text{Null}(A) = \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}.$$

Rank–nullity connection

For an $m \times n$ matrix A ,

$$\text{rank}(A) + \text{nullity}(A) = n.$$

Here n is the number of columns.

- ▶ $\text{rank}(A) = \dim(\text{Col}(A))$
- ▶ $\text{nullity}(A) = \dim(\text{Null}(A))$

Tricks and Tips

Row-reduce A :

$$\text{number of pivots} = \text{rank}(A), \quad n - \text{number of pivots} = \text{nullity}(A).$$

Example

A 2×3 matrix maps a 3D vector to a 2D vector:

$$\mathbb{R}^3 \longrightarrow \mathbb{R}^2.$$

A matrix

$$A \in \mathbb{R}^{m \times n}$$

defines

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad T(\boldsymbol{x}) = A\boldsymbol{x}.$$

7. Dot product

For $u, v \in \mathbb{R}^n$,

$$u \cdot v = u^T v = \sum_{i=1}^n u_i v_i.$$

Geometrically,

$$u \cdot v = \|u\| \|v\| \cos \theta.$$

- ▶ Positive $\Rightarrow$ acute angle.
- ▶ Zero $\Rightarrow$ perpendicular.
- ▶ Negative $\Rightarrow$ obtuse angle.

Projection

Projection of u onto v :

$$\text{proj}_v u = \frac{u \cdot v}{v \cdot v} v.$$

Duality: vectors as linear functionals

A vector $\mathbf{a}$ can define a linear functional

$$f_{\mathbf{a}}(\mathbf{x}) = \mathbf{a}^{\top} \mathbf{x}.$$

Thus the dot product converts a vector into a rule that maps vectors to scalars:

$$\mathbb{R}^n \longrightarrow \mathbb{R}.$$

Why this matters

The **dual space** consists of linear functionals. In Euclidean spaces, the dot product provides a natural correspondence between vectors and these functionals.

Machine-learning connection

A neuron's weighted sum is

$$z = \mathbf{w}^{\top} \mathbf{x} + b,$$

a linear functional plus a bias.

8. Cross product in $\mathbb{R}^3$

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix},$$

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}.$$

Properties:

$$\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u}), \quad \mathbf{u} \times \mathbf{v} \perp \mathbf{u}, \mathbf{v}.$$

Magnitude:

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta.$$

Cross product and area

The parallelogram generated by $\mathbf{u}, \mathbf{v}$ has area

$$A = \|\mathbf{u} \times \mathbf{v}\|.$$

The triangle has area

$$A_{\triangle} = \frac{1}{2} \|\mathbf{u} \times \mathbf{v}\|.$$

Quick example

$$\mathbf{u} = (1, 0, 0), \quad \mathbf{v} = (0, 2, 0)$$

gives

$$\mathbf{u} \times \mathbf{v} = (0, 0, 2).$$

So the parallelogram area is 2.

Cross products through linear transformations

For an invertible 3×3 matrix A ,

$$(A\mathbf{u}) \times (A\mathbf{v}) = \det(A) A^{-T}(\mathbf{u} \times \mathbf{v}).$$

This explains why the cross product transforms differently from an ordinary vector under a general linear transformation.

Special case: rotation

If R is a proper rotation,

$$\det(R) = 1, \quad R^{-T} = R,$$

so

$$(R\mathbf{u}) \times (R\mathbf{v}) = R(\mathbf{u} \times \mathbf{v}).$$

9. Cramer's rule

For a square system

$$A\mathbf{x} = \mathbf{b}, \quad \det(A) \neq 0,$$

the i th variable is

$$x_i = \frac{\det(A_i)}{\det(A)}$$

where A_i is obtained by replacing column i of A with $\mathbf{b}$.

For two variables

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

$$x = \frac{\begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}, \quad y = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}.$$

10. Change of basis

Suppose $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis. The coordinate vector of $\mathbf{x}$ in basis B is

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \quad \text{where} \quad \mathbf{x} = c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n.$$

Construct the basis matrix

$$P_B = [\mathbf{b}_1 \ \mathbf{b}_2 \ \dots \ \mathbf{b}_n].$$

Then

$$\boxed{\mathbf{x} = P_B [\mathbf{x}]_B}$$

and

$$\boxed{[\mathbf{x}]_B = P_B^{-1} \mathbf{x}}.$$

Change of basis between two bases

Let B and C be bases. Then

$$[\mathbf{x}]_C = P_C^{-1} P_B [\mathbf{x}]_B.$$

The matrix

$$P_{C \leftarrow B} = P_C^{-1} P_B$$

converts coordinates from B to C .

Key idea

The physical vector $\mathbf{x}$ does not change. Only its **coordinate description** changes.

11. Eigenvectors and eigenvalues

A nonzero vector $\mathbf{v}$ is an eigenvector of A if

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Here λ is the corresponding eigenvalue. Rearrange:

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

For a nonzero solution,

$$\det(A - \lambda I) = 0.$$

This equation is the **characteristic equation**.

How to find eigenvalues and eigenvectors

- 1 Compute $A - \lambda I$.
- 2 Set $\det(A - \lambda I) = 0$.
- 3 Solve for eigenvalues λ .
- 4 For each λ , solve

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

- 5 Give the eigenspace as the span of the eigenvectors.

Important

An eigenvector can never be the zero vector.

Geometric meaning of eigenvectors

Most vectors change both direction and magnitude under a transformation. An eigenvector changes only by a scalar:

$$v \mapsto \lambda v.$$

- ▶ $|\lambda| > 1$: stretching
- ▶ $0 < |\lambda| < 1$: shrinking
- ▶ $\lambda < 0$: direction reversal
- ▶ $\lambda = 1$: unchanged
- ▶ $\lambda = 0$: mapped to zero

AI connection

Eigenvectors appear in PCA, spectral methods, covariance analysis, graph algorithms and dynamical systems.

12. Vector spaces

A vector space over $\mathbb{R}$ is a set V with vector addition and scalar multiplication satisfying the vector-space axioms. Typical examples:

$\mathbb{R}^n$, polynomials, matrices, functions.

A subset $W \subseteq V$ is a subspace if:

- ❶ $\mathbf{0} \in W$,
- ❷ $\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W$,
- ❸ $\mathbf{u} \in W, c \in \mathbb{R} \Rightarrow c\mathbf{u} \in W$.

Subspaces you should recognize

For an $m \times n$ matrix A :

$$\text{Col}(A) \subseteq \mathbb{R}^m, \quad \text{Null}(A) \subseteq \mathbb{R}^n.$$

Both are vector spaces.

Column space

All possible outputs Ax .

Null space

All inputs mapped to 0.

$$\text{rank}(A) = \dim \text{Col}(A), \quad \text{nullity}(A) = \dim \text{Null}(A).$$

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}$$

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

$$A^{-1} = \frac{1}{\det A} \text{adj}(A)$$

$$\text{rank}(A) + \text{nullity}(A) = n$$

$$A\mathbf{v} = \lambda\mathbf{v}$$

$$\det(A - \lambda I) = 0$$

$$[\mathbf{x}]_B = P_B^{-1} \mathbf{x}$$

$$\mathbf{x} = P_B [\mathbf{x}]_B$$

$$x_i = \frac{\det(A_i)}{\det(A)}$$

One integrated example

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Determinant

$$\det A = 4 - 1 = 3 \neq 0.$$

Therefore A is invertible.

$$A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Eigenvalues

$$\det(A - \lambda I) = (2 - \lambda)^2 - 1$$

$$= (\lambda - 1)(\lambda - 3).$$

Hence

$$\lambda_1 = 1, \quad \lambda_2 = 3.$$